MEMORANDUM

To:	Professor Brendon Anderson, Mechanical Engineering, California Polytechnic State University, bga@calpoly.edu
From:	Gabriel Choi
Date:	April 06, 2026
Subject:	Lab 1A: Open-Loop Simulation

Introduction

The purpose of this lab was to introduce system modeling using Simulink and to analyze first-order and second-order dynamic systems. Simulations were performed for a mass-damper system, first-order systems, and second-order systems to observe their time responses to step inputs. The results demonstrate how system parameters such as gain, damping ratio, and time constants affect system behavior.

Methodology

Simulink was used to model and simulate all systems. In Exercise 1, a mass-damper system was modeled using differential equations. In Exercise 2, first-order systems were implemented using transfer function blocks. In Exercise 3, second-order systems were modeled using standard second-order transfer functions with varying damping ratios and natural frequencies. In Exercise 4, two first-order systems were connected in series to observe combined system behavior. Simulation data was exported using "To Workspace" blocks and plotted in MATLAB. Simulation parameters such as stop time and solver tolerance were adjusted to ensure accurate system responses.

Results

1A.1.1 Mass-Damper System

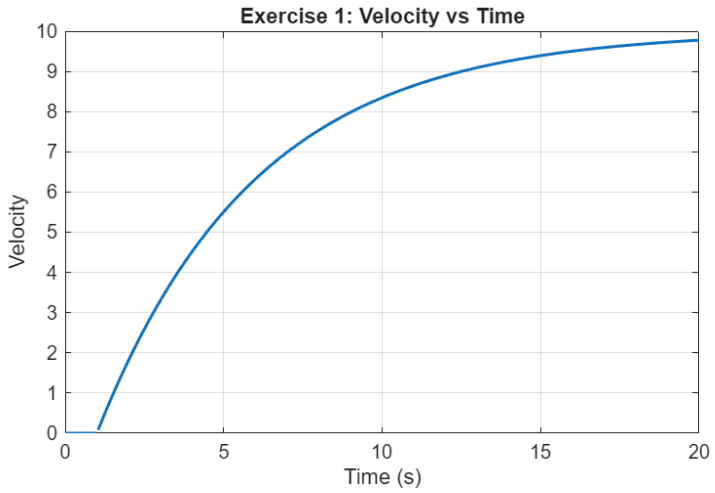


Figure 1. Velocity response of mass-damper system

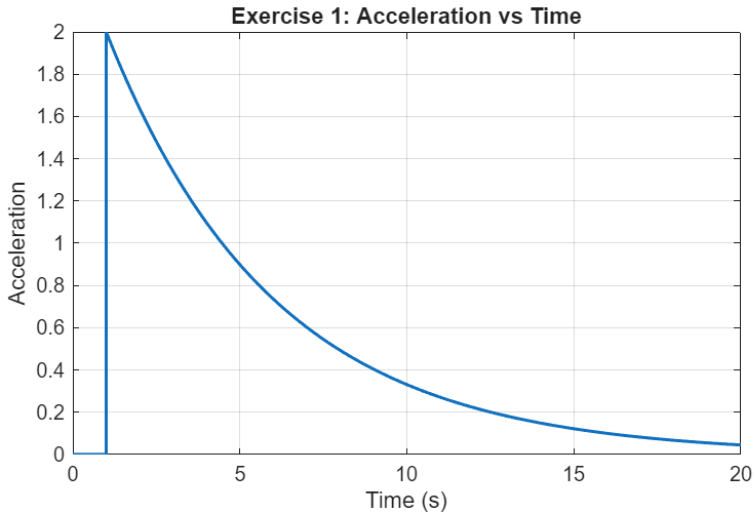


Figure 2. Acceleration response of mass-damper system

Figure 1 shows the velocity response of the mass-damper system. The velocity remains at zero until the step input is applied at $t = 1$s, after which it increases smoothly and approaches a steady-state value. Figure 2 shows the acceleration response. The acceleration exhibits a sudden increase at the step input and then decays toward zero as the system reaches steady state.

1A.2.1 First-Order Systems

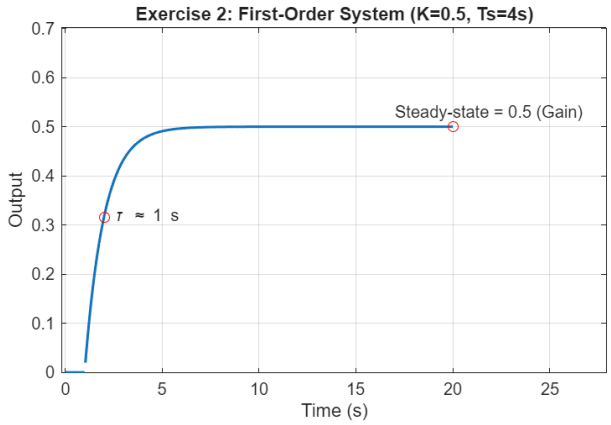


Figure 3. First-order system (K=0.5, Ts=4s)

The steady-state value of 0.5 represents the system gain, and the time constant is approximately 1 second, defined as the time at which the response reaches 63.2% of its final value.

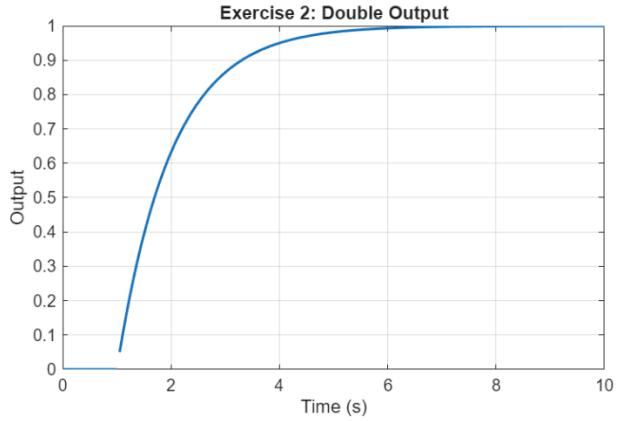


Figure 4. First-order system with doubled output

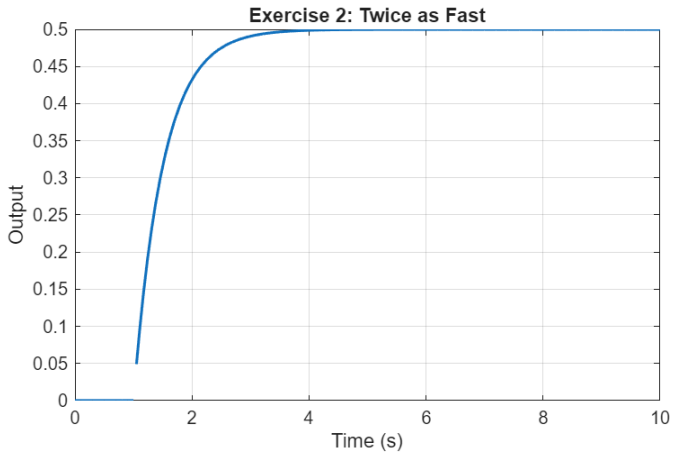


Figure 5. First-order system with reduced time constant

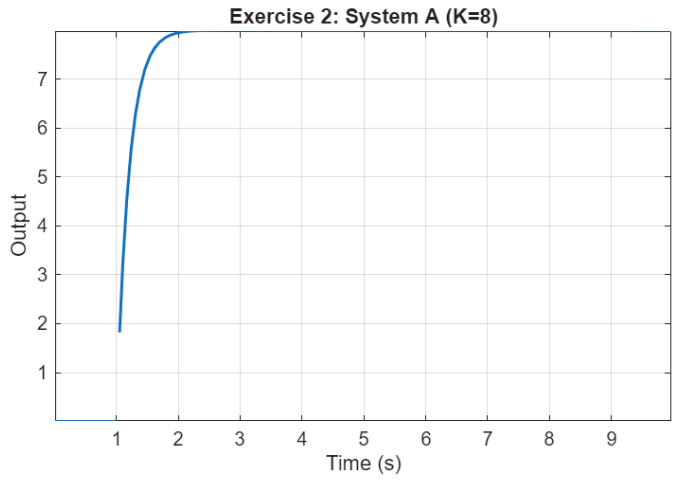


Figure 6. First-order system with high gain (System A)

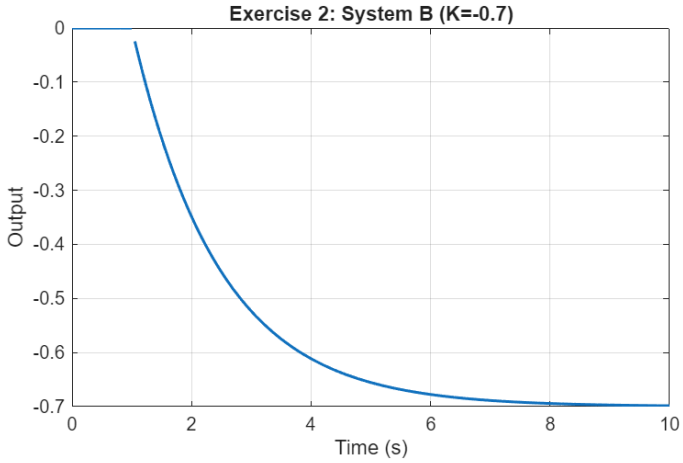


Figure 7. First-order system with negative gain (System B)

Figure 3 shows the response of the original first-order system. The output approaches a steady-state value of approximately 0.5. Figure 4 shows the system with doubled output, which reaches a steady-state value of approximately 1.0. Figure 5 shows the system with a reduced time constant, resulting in a faster response while maintaining the same steady-state value.

Figure 6 shows System A, which has a higher gain and faster response, reaching a steady-state value near 8. Figure 7 shows System B, which has a negative gain and responds in the negative direction, settling near -0.7.

1A.3.1 Second-Order Systems

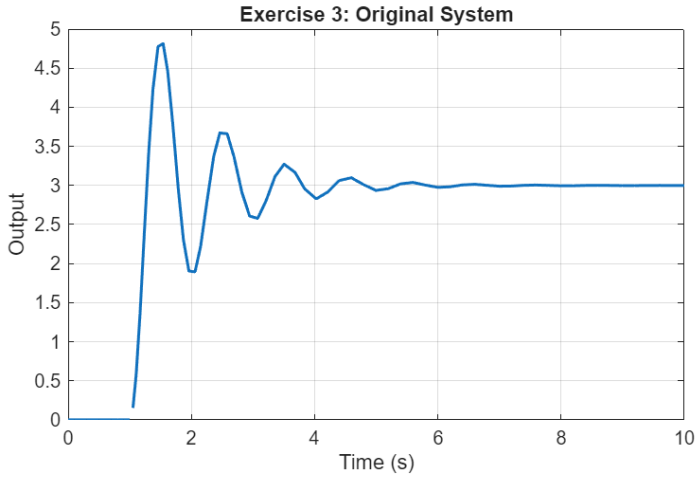


Figure 8. Original second-order system response

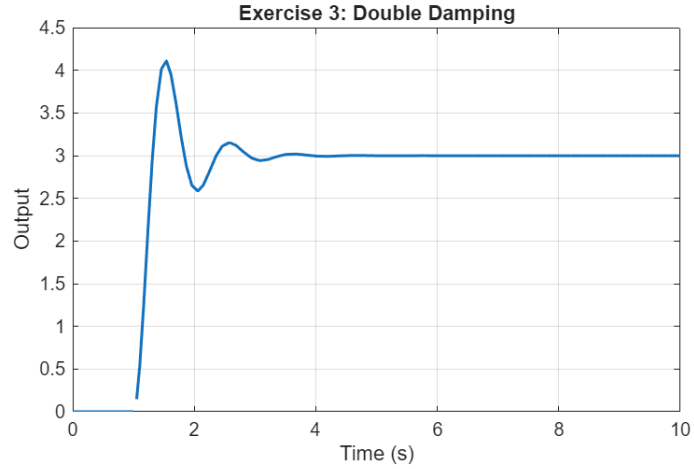


Figure 9. Second-order system with increased damping

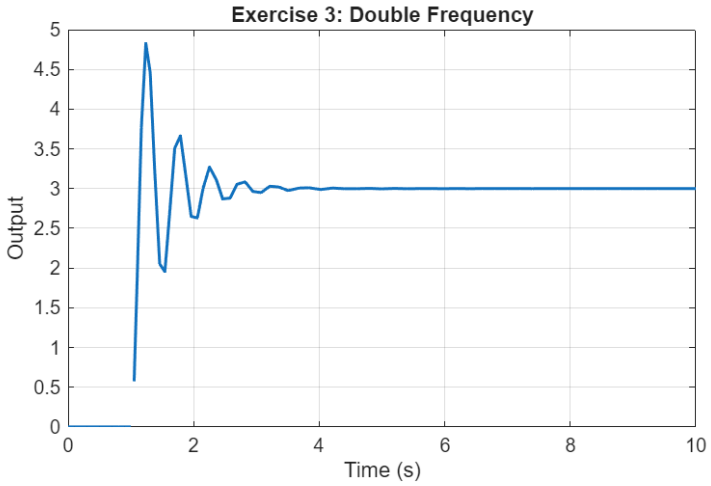


Figure 10. Second-order system with increased natural frequency

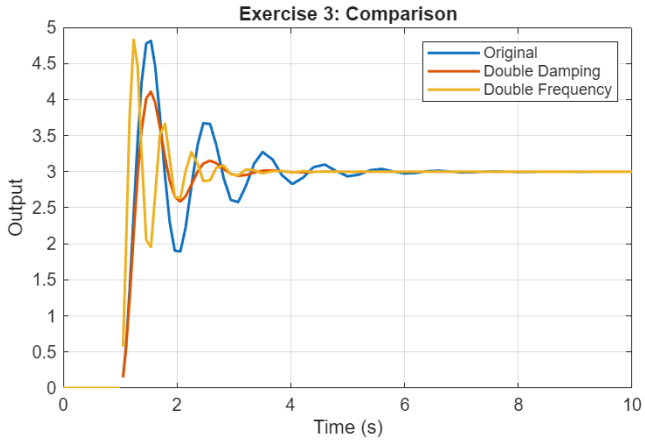


Figure 11. Comparison of second-order systems

Figure 8 shows the response of the original second-order system. The system exhibits oscillations and overshoot before settling at a steady-state value of approximately 3. Figure 9 shows the system with increased damping, which reduces overshoot and oscillations while maintaining the same steady-state value. Figure 10 shows the system with increased natural frequency, which results in a faster response with more rapid oscillations.

1A.3.2 Combined First-Order Systems

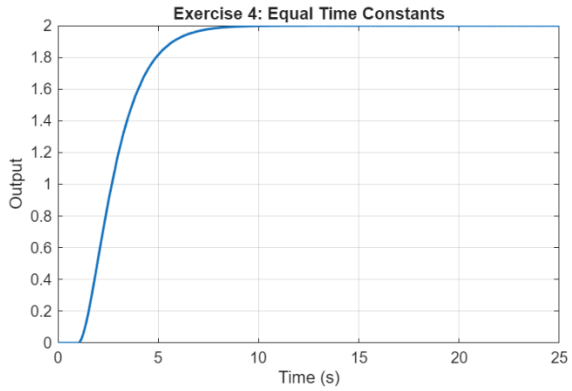


Figure 12. Equal time constant systems (series)

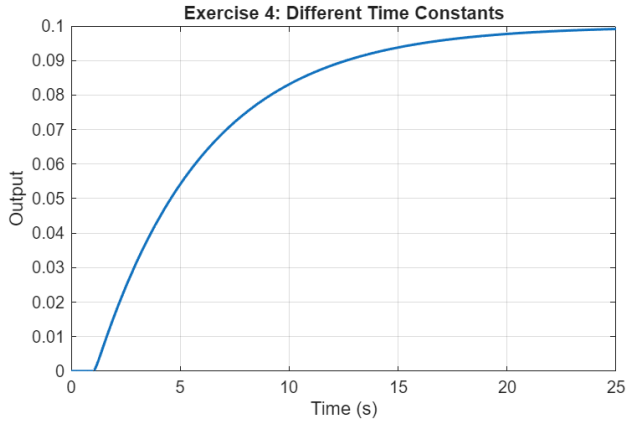


Figure 13. Different time constant systems

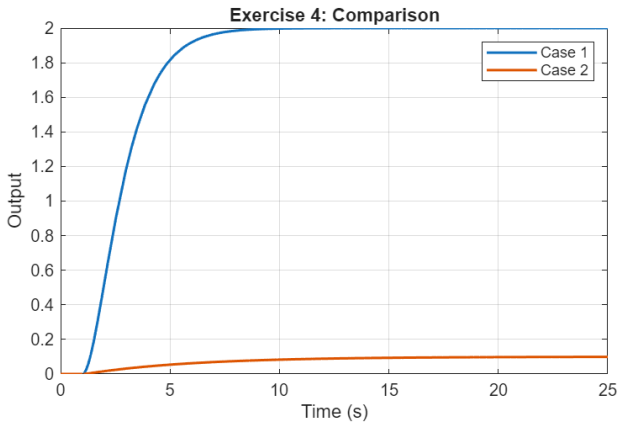


Figure 14. Comparison of series systems

Figure 12 shows the response of two identical first-order systems in series. The response is slower than a single first-order system and resembles a second-order response without oscillation. Figure 13 shows the response of two systems with significantly different time constants. The response is dominated by the slower system and behaves similarly to a first-order system with a large time constant.

Discussion

1A.2.1 First-Order Systems

Question: Does the system behave as expected?
Yes, the system exhibits the expected exponential response of a first-order system, approaching a steady-state value without oscillation.

Question: How long does it take to reach 63.2% of its final value?
The system reaches 63.2% of its final value at one time constant. For this system, the time constant is 1 second, so the response reaches 63.2% at approximately 1 second.

Question: What is the size of the output step?
The output step size is equal to the system gain multiplied by the input step. For the original system, the output step is approximately 0.5.

Question: How could you change the system to produce the same output but twice as fast?
The system can be made twice as fast by reducing the time constant by half.

Question: How could you double the output?
The output can be doubled by doubling the system gain.

Question: Did your modified systems behave as expected?
Yes, increasing the gain increased the steady-state value, and reducing the time constant resulted in a faster response.

1A.3.1 Second-Order Systems

Question: What effect does doubling the damping coefficient have?
Doubling the damping coefficient reduces the overshoot and oscillations in the system, resulting in a smoother response.

Question: What effect does doubling the oscillation frequency have?
Doubling the natural frequency increases the speed of the response and causes oscillations to occur more rapidly while maintaining the same steady-state value.

1A.3.2 Combined First-Order Systems

Question: Does the system behave like a first-order system?
When two identical first-order systems are combined, the system behaves as a second-order system, although it does not exhibit oscillatory behavior.

Question: Do two first-order systems in series constitute a second-order system?
Yes, two first-order systems in series form a second-order system.

Question: What role does each subsystem play in determining the response?
In systems with different time constants, the slower subsystem dominates the overall response, while the faster subsystem has minimal effect on the long-term behavior.

Appendix

```
t = out.tout;
v = out.vel;

figure;
plot(t, v, 'LineWidth', 1.5);
grid on;
title('Velocity vs Time');
xlabel('Time (s)');
ylabel('Velocity');
```

Figure 15. Example MATLAB Plotting Code Snippet

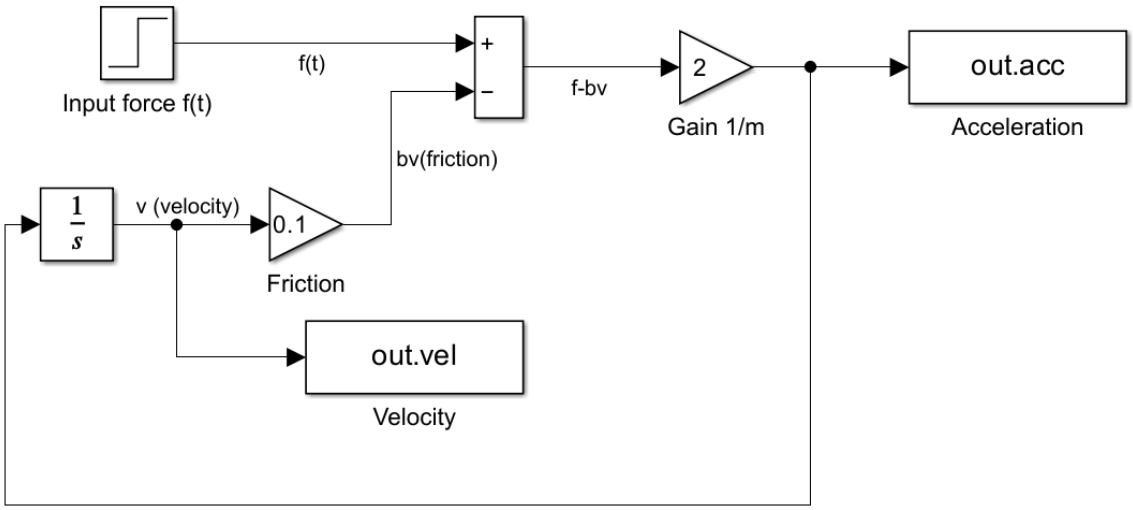


Figure 16. Exercise 1 Simulink

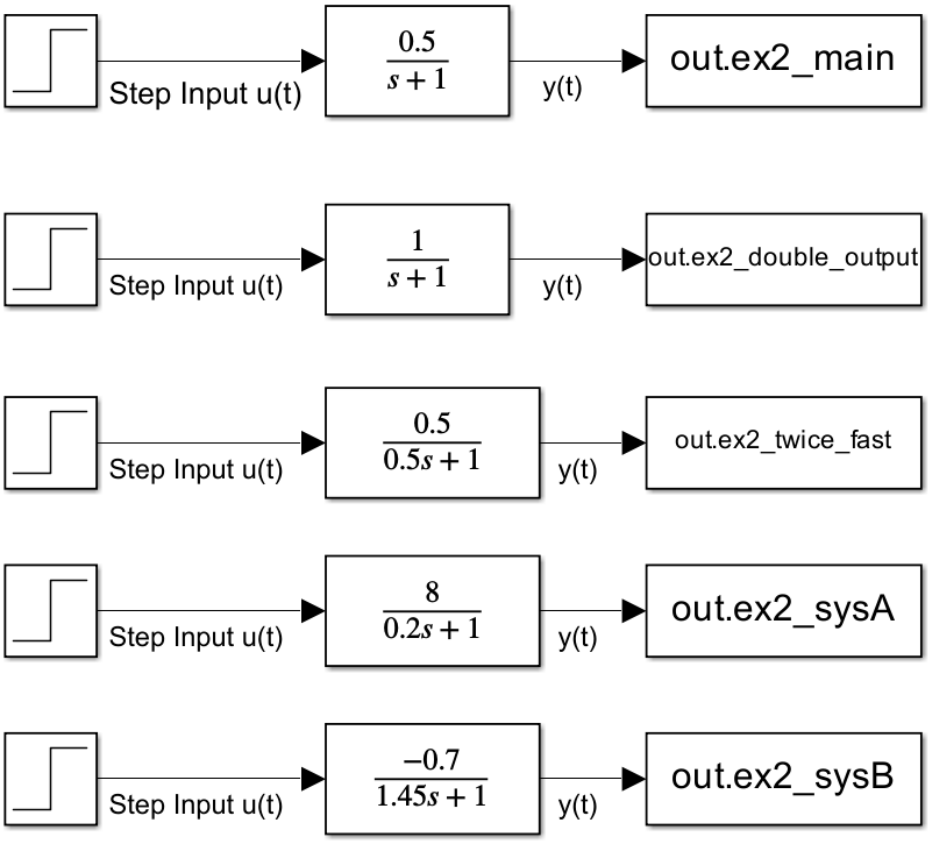


Figure 17. Exercise 2 Simulink

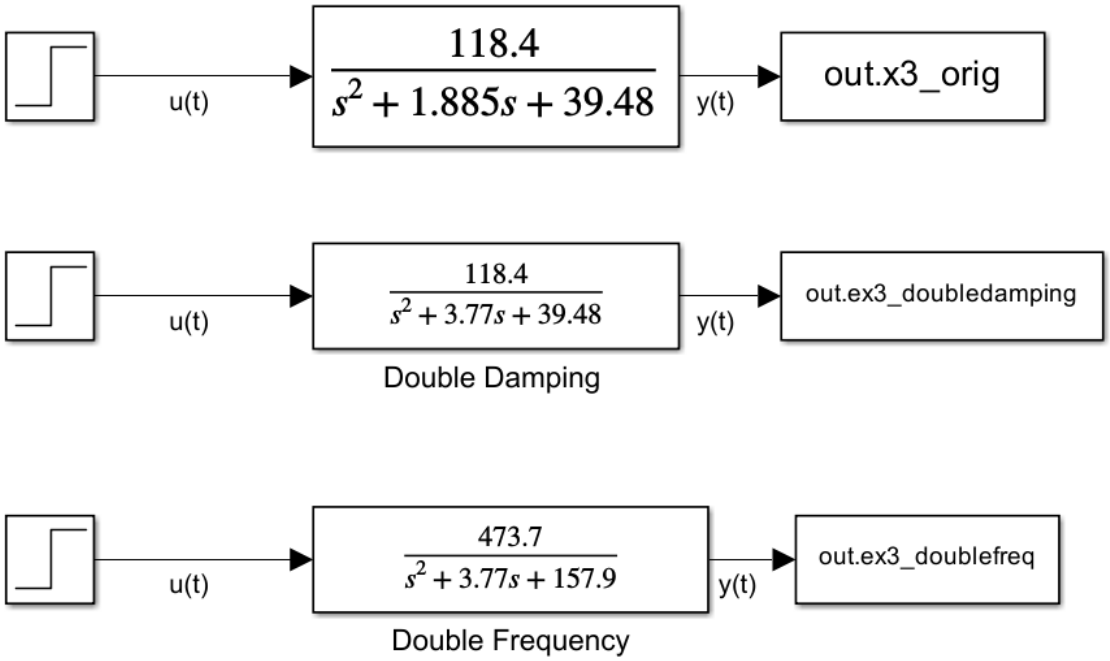


Figure 18. Exercise 3 Simulink

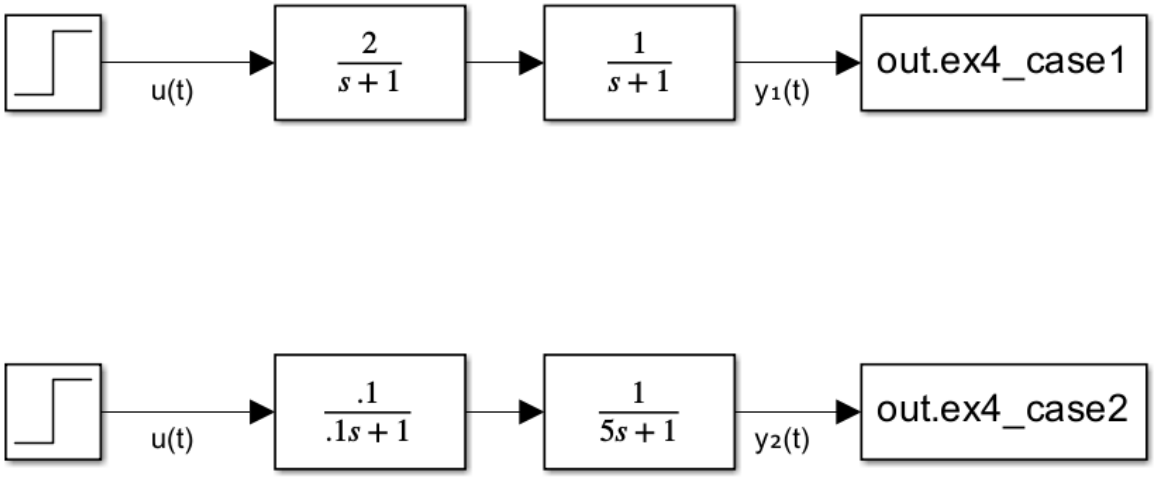


Figure 19. Exercise 4 Simulink